



## Differentiability for functions of one variable

### Definition

Let  $f$  be a function defined on an open interval around  $a$ . Then  $f$  is **differentiable at  $a$**  if  $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  exists.

For a function  $f(x)$  its **derivative** function,  $f'(x)$  or  $\frac{df}{dx}(x)$  is

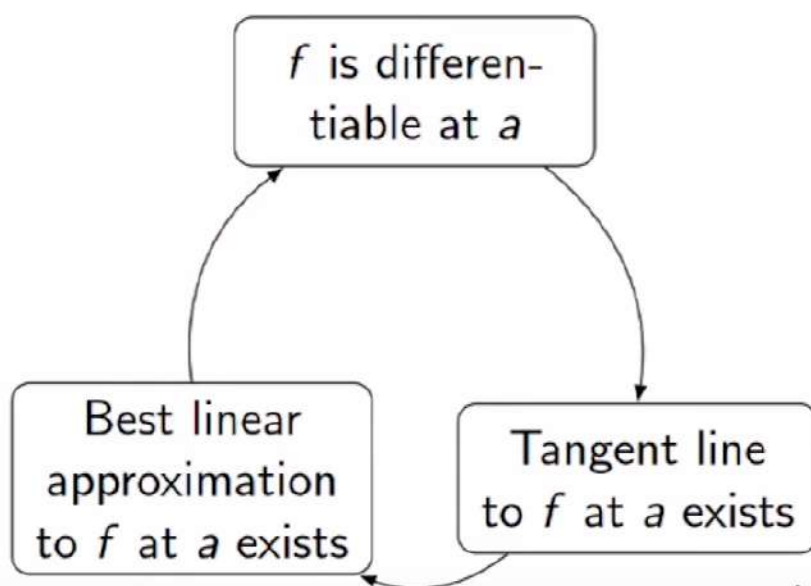
$$f'(x) = \frac{df}{dx}(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Its domain consists of those points at which the function  $f(x)$  is differentiable.

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**Theorem:** If  $f$  is differentiable at  $a$ , then it is continuous at  $a$ .

# Differentiability, tangents and best linear approximation



$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - f'(a)h}{h} = 0.$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - \boxed{(f(a) + f'(a)h)}}{h} = 0.$$

$$y = f(a) + f'(a)(x-a)$$

$$L_f(x) = f(a) + f'(a)(x-a)$$

$$f(x) \approx L_f(x)$$

when  $x$  is near  $a$ .

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# Differentiability for scalar-valued multivariable functions

## Definition

Let  $f$  be a scalar-valued multivariable function defined on a domain  $D$  in  $\mathbb{R}^n$  containing an open ball around a point  $\underline{a}$ .

Then  $f$  is **differentiable at  $\underline{a}$**  if

$$\lim_{\underline{h} \rightarrow 0} \frac{f(\underline{a} + \underline{h}) - f(\underline{a}) - \underline{h} \cdot \nabla f(\underline{a})}{\|\underline{h}\|} = 0.$$

2-args.  $\underline{h} = (h_1, h_2)$

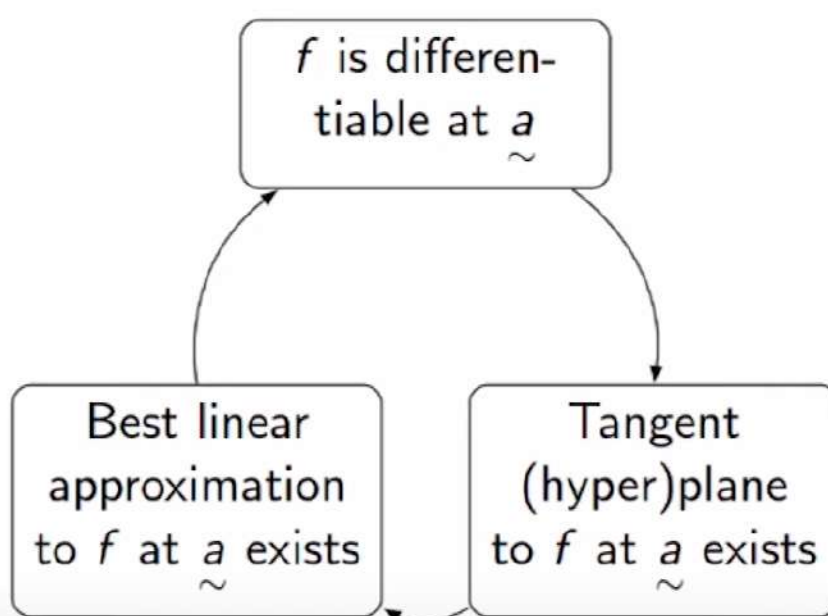
$$\lim_{(h_1, h_2) \rightarrow (a, b)} \frac{f((a, b) + (h_1, h_2)) - f(a, b) - h_1 f_x(a, b) - h_2 f_y(a, b)}{\sqrt{h_1^2 + h_2^2}} = 0$$

$$f(\underline{a} + \underline{h}) - \{f(\underline{a}) + \underline{h} \cdot \nabla f(\underline{a})\}$$

$$= f(\underline{a}) + \nabla f(\underline{a}) \cdot (\underline{x} - \underline{a})$$



# Differentiability, tangents and best linear approximation



Fact : If  $f$  is differentiable at  $a$ , then it is continuous at  $a$ .

$\frac{\partial f}{\partial x_i}(x)$  is cts. in a ball around  $a$ .